

Total No. of Questions : 8]

SEAT No. :

PC2805

[6352]-29

[Total No. of Pages : 3

S.E. (Electronics/Electronics & Telecommunication)

DATA STRUCTURES

(2019 Pattern) (Semester - III) (204184)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of calculator is allowed.
- 5) Assume suitable data if necessary.

- Q1)** a) Compare Stack and Queue. What are the advantages of circular queue over liner queue? [6]
- b) Write a function PUSH and POP in 'C' for stack using linked list. [6]
- c) What are the applications of Queue? Explain two applications in detail. [5]

OR

- Q2)** a) Write a short note on circular queue. Compare it with linear queue. [5]
- b) Convert the following prefix expression into infix form. Show all the steps and stack contents: [6]
- *-A/BC-/AKL
- c) Write ADD and DETETE function in 'C' for Queue using array. [6]

- Q3)** a) Compare array and linked list. [5]
- b) Write a 'C' function to delete a number from singly linked list. [6]
- c) Explain doubly linked list (DLL). What are the advantages of DLL over SLL. [6]

OR

P.T.O.

Q4) a) Draw and explain circular linked list. State the limitations of single linked list. [5]

b) Write a 'C' function to insert a number at end in to the singly linked list. [6]

c) Differentiate singly linked list and doubly linked list. [6]

Q5) a) Construct Binary search tree for the following : [6]

MAR, OCT, JAN, APR, NOV, FEB, MAY, DEC, JUN, AUG, JUL, SEP

b) Write a pseudo code to search an element in binary search tree using arrays. [6]

c) Explain with suitable example how binary tree can be represented using: [6]

i) Array

ii) Linked List

OR

Q6) a) Define BST? Create a BST for the following data: [6]

14, 15, 4, 9, 7, 18, 3, 5, 7.

b) Define binary tree. Name and explain with suitable example the following terms [6]

i) Root node

ii) Left sub tree and right sub tree

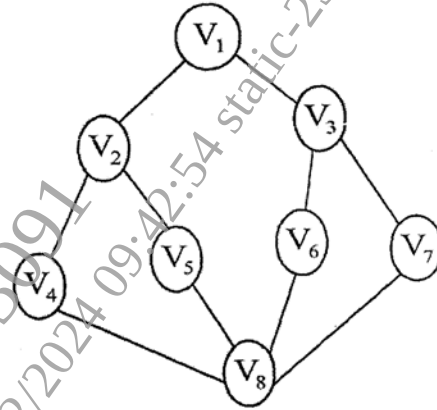
iii) Depth of tree

c) Construct the binary search tree from the following elements: [6]

15, 4, 16, 8, 2, 18, 14

Also show preorder, inorder and postorder traversal for the same.

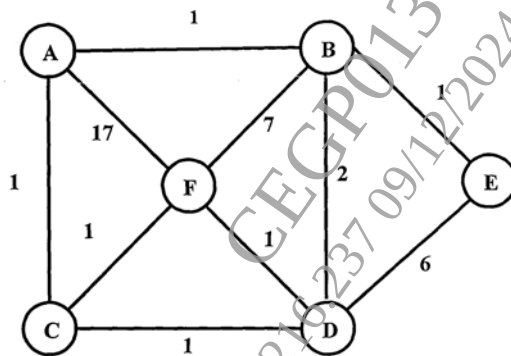
- Q7) a)** Draw adjacency list and adjacency matrix for the following graph: [6]



- b) What is MST? Explain with suitable example Kruskal's Algorithm to find out MST. [6]
- c) Define DFS and BFS graph with example. [6]

OR

- Q8) a)** Explain Kruskal algorithm? Find the minimum spanning tree for below figure. Using Kruskal's Algorithm. [6]



- b) Explain Dijkstra's algorithm with example. [6]
- c) Explain with suitable example the techniques to represent a Graph. [6]

Note: consider graph of minimum 6 vertices

x

x

x